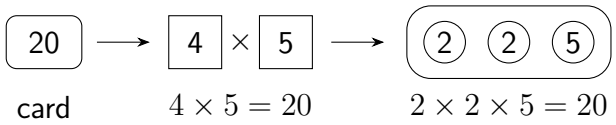
A number card can split into two whole numbers greater than 1 that multiply to its value. These numbers are its factors. Keep splitting until every card is prime: a number greater than 1 with no factors except 1 and itself. The finished cards make a prime bag. Multiply its tokens to find its value.



Token order does not matter. Equal tokens are interchangeable.

Problem 1: For each card, find two different splitting routes. Can different routes finish with different prime bags?

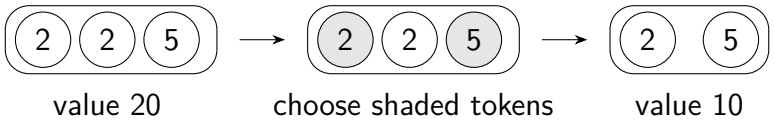
12

18

24

36

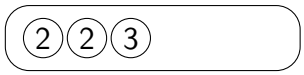
Choose some tokens from a bag, using each token at most once. The chosen tokens form a bag. You may choose all of them or none. An empty bag has value 1. A divisor of a whole number divides it with no remainder.



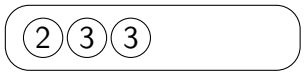
value 20 choose shaded tokens value 10

$2 \times 5 = 10$ $20 \div 10 = 2$

Problem 2: For each bag, find every different value you can make by choosing tokens. Does your collection contain every divisor of the full bag's value?

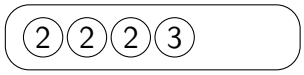


value 12

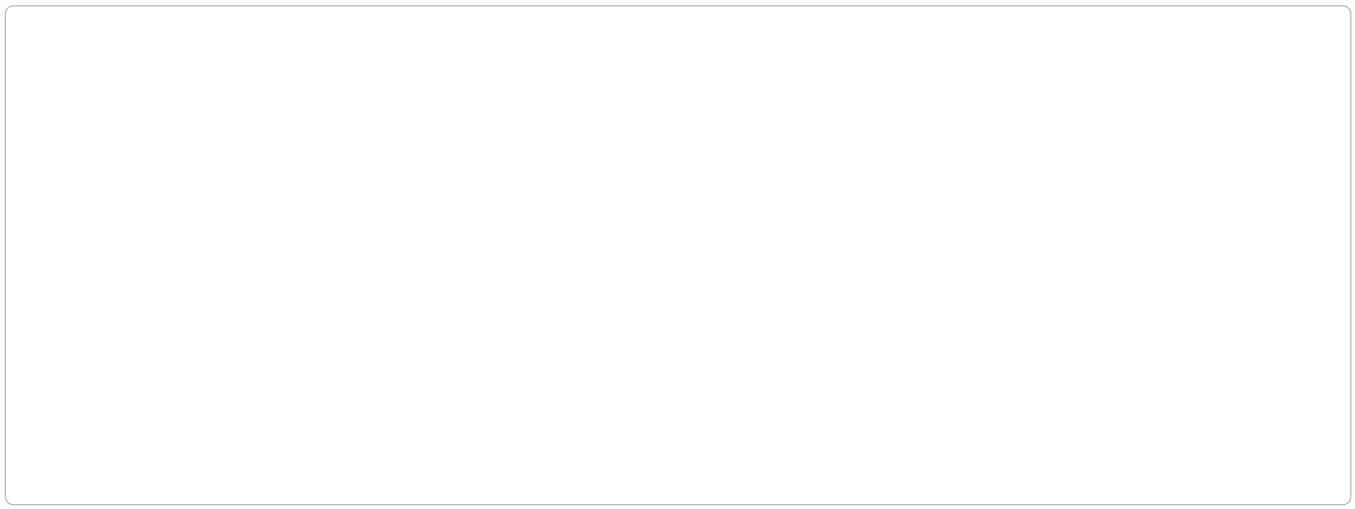


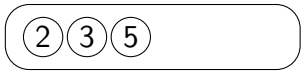
value 18

Problem 3: These bags have different numbers of tokens. Which can make more different values by choosing tokens? Find the complete collection for each bag.



value 24





value 30

